

LABSHEET - 3**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# 1. Load image

# IMPORTANT: r before the path prevents Windows path errors
image_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"
img = cv2.imread(image_path)

if img is None:
    print("Error: Image not found!")
    print("Please check the image path and filename.")
    exit()

# Convert BGR to RGB for displaying with Matplotlib
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully!")
print("Image size:", gray.shape)
print("Himani Chaudhary")
print("CU240250967")
```

```
Image loaded successfully!
Image size: (3584, 5376)
Himani Chaudhary
CU240250967
```

```
# 2. Gaussian Filter - Low Pass Filter
# -----

gaussian = cv2.GaussianBlur(gray, (5, 5), 0)
print("Himani Chaudhary")
print("CU240250967")
```

```
Himani Chaudhary
CU240250967
```

```
# 3. Median Filter - Low Pass Filter
# -----

median = cv2.medianBlur(gray, 5)
print("Himani Chaudhary")
print("CU240250967")
```

```
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```

```
# 4. Average / Mean Filter - Low Pass Filter
# -----

average = cv2.blur(gray, (5, 5))
print("Himani Chaudhary")
print("CU240250967")
```

```
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```

```
# 5. Laplacian Filter - High Pass Filter
# -----

laplacian = cv2.Laplacian(gray, cv2.CV_64F)

# Convert to unsigned 8-bit
laplacian = cv2.convertScaleAbs(laplacian)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 6. Sobel Edge Detection
# -----

# Horizontal edges
sobel_x = cv2.Sobel(
    gray,
    cv2.CV_64F,
    1,
    0,
    ksize=3
)

# Vertical edges
sobel_y = cv2.Sobel(
    gray,
    cv2.CV_64F,
    0,
    1,
    ksize=3
)

# Convert to 8-bit
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Combined Sobel
sobel_combined = cv2.addWeighted(
    sobel_x,
    0.5,
    sobel_y,
    0.5,
    0
)

print("Himani Chaudhary")
print("CU240250967")
```

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```
# 7. Display Original and Filtered Images
# -----

plt.figure(figsize=(15, 10))

plt.subplot(3, 3, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(3, 3, 2)
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Filter")
plt.axis("off")

plt.subplot(3, 3, 3)
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")

plt.subplot(3, 3, 4)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.axis("off")

plt.subplot(3, 3, 5)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")
```

```

plt.subplot(3, 3, 6)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X - Horizontal")
plt.axis("off")

plt.subplot(3, 3, 7)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y - Vertical")
plt.axis("off")

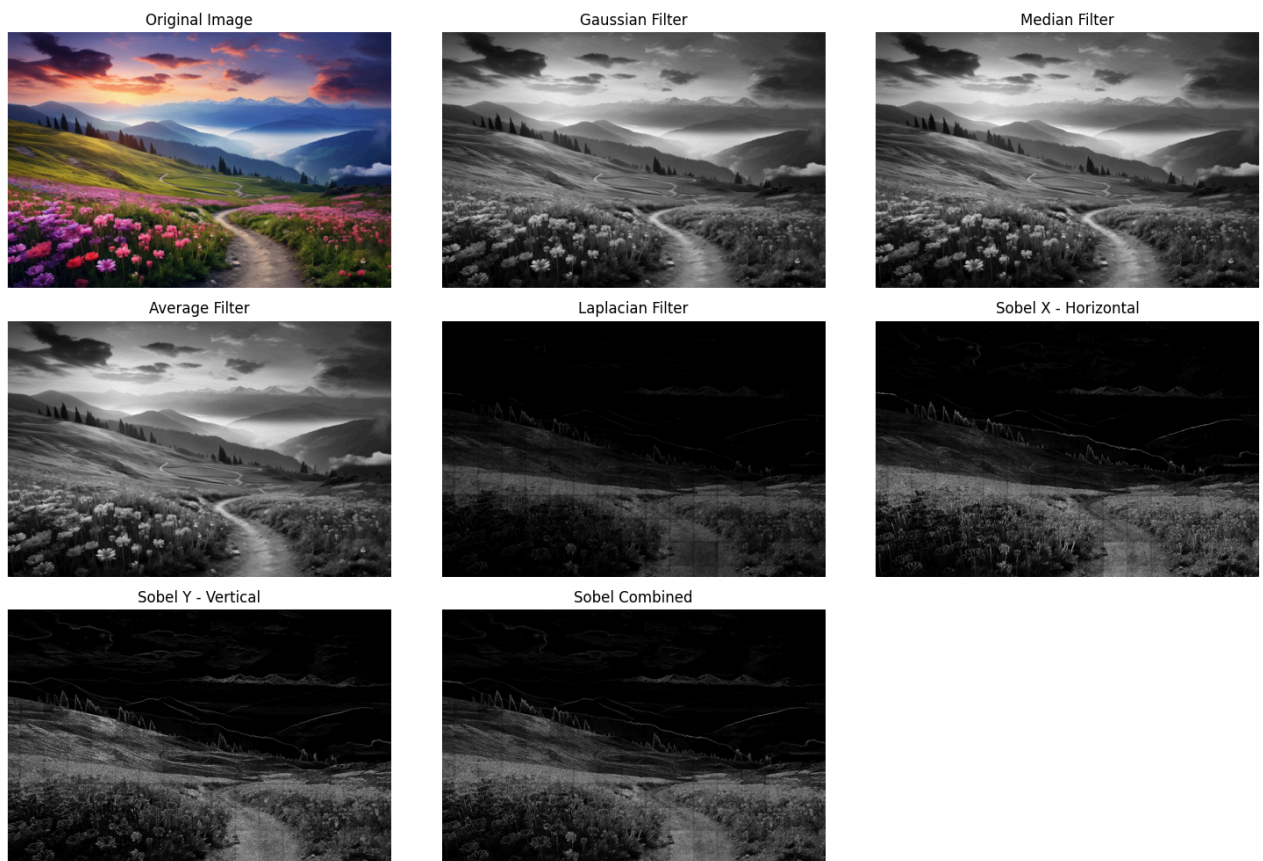
plt.subplot(3, 3, 8)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Sobel Combined")
plt.axis("off")

plt.tight_layout()

plt.savefig(
    "spatial_filtering_result.jpg",
    dpi=300
)

plt.show()
print("Himani Chaudhary")
print("CU240250967")

```



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```

# 8. Save Output Images
# -----

cv2.imwrite("gaussian.jpg", gaussian)
cv2.imwrite("median.jpg", median)
cv2.imwrite("average.jpg", average)
cv2.imwrite("laplacian.jpg", laplacian)
cv2.imwrite("sobel_x.jpg", sobel_x)
cv2.imwrite("sobel_y.jpg", sobel_y)
cv2.imwrite("sobel_combined.jpg", sobel_combined)

print("\nOutput images saved successfully!")
print("Himani Chaudhary")
print("CU240250967")

```

Output images saved successfully!
 Himani Chaudhary
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```
# 9. Display Observations
# -----

print("\n===== OBSERVATIONS =====")

print("""
1. Gaussian Filter:
  - Smooths the image.
  - Reduces random noise.
  - Preserves important structures better than average filtering.

2. Median Filter:
  - Very effective for salt-and-pepper noise.
  - Preserves edges better than average filtering.

3. Average Filter:
  - Smooths the image.
  - Reduces noise.
  - Can blur edges and fine details.

4. Laplacian Filter:
  - Detects rapid changes in intensity.
  - Enhances edges and fine details.
  - It is a high-pass filtering technique.

5. Sobel Filter:
  - Detects image edges.
  - Sobel X detects changes in the horizontal direction.
  - Sobel Y detects changes in the vertical direction.
  - Combined Sobel shows edges in both directions.
""")
print("Himani Chaudhary")
print("CU240250967")
```

```
===== OBSERVATIONS =====

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```

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```
# 10. Simple Comparison
# -----

print("===== FILTER COMPARISON =====")

print("""
Gaussian  -> Good smoothing and noise reduction
Median    -> Best for salt-and-pepper noise
Average   -> Simple smoothing but may blur edges
Laplacian -> Edge and detail enhancement
Sobel     -> Edge detection in X and Y directions
""")
```

```
print("Experiment 3 completed successfully!")  
print("Himani Chaudhary")  
print("CU240250967")
```

===== FILTER COMPARISON =====

Gaussian -> Good smoothing and noise reduction
Median -> Best for salt-and-pepper noise
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Sobel -> Edge detection in X and Y directions

Experiment 3 completed successfully!
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